

Alexander Tsai

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EDUCATION

University of California, Berkeley

Global Computing and Data Science Program

- Courses: Intro to Machine Learning, Intro to AI, Computer Vision and Computational Photography

Berkeley, CA

Aug 2027 – Dec 2027

Singapore Management University

B.S. in Human-Computer Interactions and Data Science

- Dean's List AY2024-2026

Singapore

Aug 2023 – May 2027

EXPERIENCE

Quantitative Research Intern

ExodusPoint Capital Management

- Trained **transformer models in PyTorch** to forecast market levels from cross-asset data and market events.
- Evaluated scaling across **30M–500M parameters**, observing a monotonic decrease in MSE as model size increased.
- Integrated **learned embeddings** from the transformer models into existing strategies, improving **out-of-sample Sharpe by 0.24**.

May 2026 – Aug 2026

New York, NY

AI Research Intern

TikTok

- Developed an **AI-edited image detector** that identifies edits through deviations from real-image frequency-domain patterns, achieving **97% precision / 80% recall** on internal benchmarks and **66% recall** on unseen generators.
- Curated a **novel dataset of 400K** AI-edited images using multiple image generators for fine-tuning and evaluation.
- Visualised **patch-level attention and Fourier spectra** to localise edited regions and interpret detector predictions.

Jan 2026 – May 2026

San Jose, CA

Machine Learning Engineer Intern

TikTok

- Built and deployed a **multimodal moderation pipeline** using Qwen3-VL, vLLM, and XGBoost, screening **4K+ accounts daily**.
- Fine-tuned Qwen3-VL using **GRPO reinforcement learning** on a curated dataset, improving video feature extraction accuracy by 12%.
- Increased **inference throughput by 5.35x** on an H100 GPU using vLLM, batching, asynchronous inference, and multithreading.

Jul 2025 – Dec 2025

Singapore

Data Science Intern

Tradeweb

- Built an XGBoost model to predict transaction costs using 5 years of historical trade data totalling **\$4.7T** in notional volume.
- Reduced RMSE by **21%** over baseline via **hyperparameter optimisation** and SHAP-guided feature engineering.

May 2025 – Jul 2025

New York, NY

RESEARCH

Research Fellow

Reach Alliance

- Developed a multilingual **RAG** chatbot using **LangChain** and Firestore to broaden financial literacy access for migrant workers; findings **published** and presented at an annual University of Toronto conference.
- Presented research as a **TEDx** speaker: *Behavioural Sciences and Conversational AI*.

May 2024 – May 2025

Singapore

LEADERSHIP

AI For Youth Instructor

Sustainable Learning Lab 2

- Instructed 30 students under **Intel's AI for Youth Program**, teaching Python, statistics, and neural networks.
- Guided 6 teams on capstone projects, including computer vision-based attendance systems and emotional-support chatbots.

Aug 2024 – Dec 2024

Singapore

SKILLS

- **Programming:** Python, Bash, R, SQL
- **ML Libraries:** PyTorch, vLLM, Transformers, scikit-learn, OpenCV, LangChain
- **Infrastructure & Data:** Docker, Kubernetes, Spark, Polars, Pandas, NumPy, Git
- **Interests:** Drums, Neovim, Snowboarding